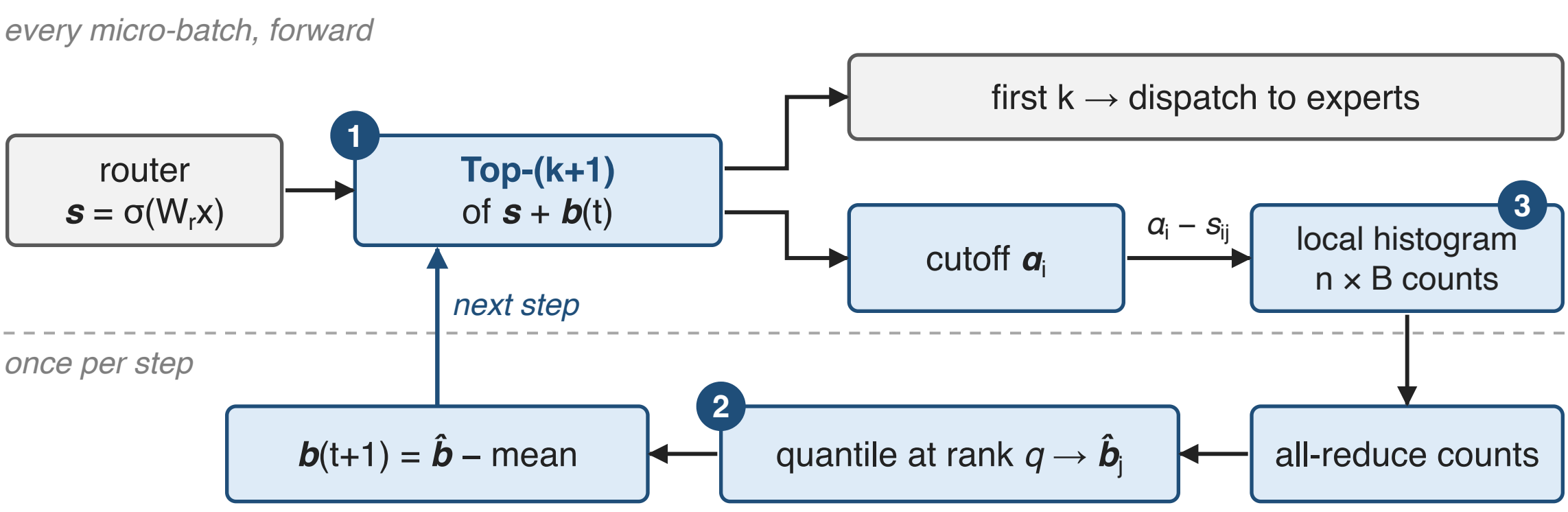

every micro-batch, forward
router
s = σ(W_r x)
1
Top-(k+1)
of s + b(t)
first k → dispatch to experts
cutoff a_i
a_i − s_ij
3
local histogram
n × B counts
next step
once per step
b(t+1) = b̂ − mean
2
quantile at rank q → b̂_j
all-reduce counts